

CPU baselines, RTL simulation, and RTL latency

Python integer kernel

123621.090 ms

MEASURED_NOW

Exact C++ native CPU kernel

1777.700 ms

MEASURED_NOW

Verilator-generated RTL simulation

2456.708 ms

MEASURED_NOW

Pure RTL

54.012600 ms

CYCLE_DERIVED

Source: [benchmarks/accelerator_benefit/figures/01_cpu_vs_rtl_latency_source.csv](#)

Limitation: Exact C++ is the native CPU baseline; Verilator is separately labeled RTL simulation runtime.